

# Generate Audit Export

## Use Case Name

Generate Audit Export

## Primary Actor

Compliance Officer / Security Auditor

## Secondary Actors

- Audit Log Service (system component that stores and serves audit records)
- Integrity Verification Service (validates SHA-256 hashes)

## Trigger

The user initiates an audit export request from the audit console (e.g., for a compliance review, incident investigation, or scheduled reporting requirement).

## Preconditions

- P1: Audit logs exist in the system and are retrievable.
- P2: The user is authenticated and authorized (role: Compliance Officer or Security Auditor) to generate audit exports.
- P3: Each audit record contains an associated SHA-256 integrity hash computed at write time.
- P4: The audit export subsystem is available and operational.

## Postconditions

### Success:

- S1: The requested audit records are exported in the selected format.
- S2: The export includes SHA-256 integrity verification information for every included record.
- S3: An audit trail entry is created logging that an export was generated, by whom, and with what filter criteria (export-of-the-export-action, for non-repudiation).

### **Partial Success (with exclusions):**

- PS1: The export is generated successfully but excludes one or more records that failed integrity verification.
- PS2: A discrepancy report accompanies the export listing excluded record IDs and the reason for exclusion.

### **Failure:**

- F1: No export is generated; the user is informed of the failure reason.

## **Main Success Scenario**

1. User selects "Generate Audit Export" from the audit console.
2. System displays available audit records with filter options (date range, event type, actor, severity, system component, etc.).
3. User selects the required records or applies filters and confirms the request.
4. System retrieves the matching audit records from the audit log store.
5. System retrieves the stored SHA-256 integrity hash for each record.
6. System recomputes each record's hash and compares it against the stored value to verify integrity.
7. System compiles the verified records and their integrity hashes into the export package.
8. System generates the audit export in the requested format (e.g., CSV, JSON, PDF).
9. System logs the export generation event (user, timestamp, filter criteria, record count) as its own auditable action.
10. System provides the completed export to the user for download.
11. Use case ends successfully.

## **Alternate Flow A: Integrity Verification Failure**

- A1 (at step 6): System detects that a record's recomputed hash does not match the stored hash, or that the hash is missing.
- A2: System marks the affected record as "integrity-invalid."
- A3: System excludes the invalid record from the export package.
- A4: System adds the record ID and failure reason to a discrepancy report.
- A5: System continues processing remaining records (resumes at step 6 for the next record).
- A6: At step 10, system delivers the export together with the discrepancy report and a clear notice that the export is incomplete.
- A7: User may review the flagged record(s) via a separate investigation flow and retry the export once resolved.
- Resumes at step 11.

## Alternate Flow B: No Records Match Filter Criteria

- B1 (at step 4): No audit records match the selected filters.
- B2: System informs the user that no records were found and suggests adjusting the filter criteria.
- B3: User may revise filters and return to step 3, or cancel the use case.

## Exception Flows

- E1: Audit Log Service unavailable at step 4 → System aborts the export, notifies the user of a temporary system error, and logs the failed attempt. User may retry later.
- E2: User's authorization expires or is revoked mid-process → System halts the export and denies delivery; the partially generated export (if any) is discarded, not cached, and not delivered.
- E3: Export generation fails after retrieval (e.g., formatting/storage error at step 8) → System retains no partial output accessible to the user, logs the failure, and prompts the user to retry.

## Business Rules

- BR1: Only users with the Compliance Officer or Security Auditor role (or an equivalent explicitly granted permission) may generate audit exports.
- BR2: Every exported record must carry its SHA-256 integrity hash; an export must never silently include a record whose integrity could not be verified.
- BR3: Any exclusion due to integrity failure must be disclosed to the user in the export deliverable itself, not only in a separate notification.
- BR4: The act of generating an audit export is itself logged as an audit event (for chain-of-custody and non-repudiation purposes).
- BR5: Exported files must not be modifiable in a way that invalidates their embedded integrity hashes without detection (e.g., export file itself may be signed or hashed as a whole).

## Non-Functional Requirements

- NFR1 (Security): Export transport and storage must use encryption in transit and, if persisted, at rest.
- NFR2 (Performance): Export generation for up to [N] records should complete within [X] seconds/minutes; for larger sets, an asynchronous/background export with notification on completion should be supported.
- NFR3 (Auditability): The export generation log entry (step 9) must be immutable and itself protected by integrity hashing.

- NFR4 (Usability): Filter selection UI should support saved/reusable filter presets for recurring compliance reports.
- NFR5 (Reliability): Integrity verification (step 6) must process each record independently so a single failure does not abort the entire export.

## **Related Use Cases**

- UC-01: Investigate Flagged Audit Record (follow-on from Alternate Flow A)
- UC-02: Manage Audit Retention Policy
- UC-03: View Audit Log (read-only browsing, precursor to export)